frontiers | Frontiers in Sustainable Food Systems

TYPE Original Research
PUBLISHED 05 January 2026
DOI 10.3389/fsufs.2025.1669746

Check for updates

OPEN ACCESS

EDITED BY
Stefano Rinaldi,
University of Brescia, Italy

REVIEWED BY
Gandhi Pawitan,
Parahyangan Catholic University, Indonesia
Gonda Yumitro,
Universitas Muhammadiyah Malang, Indonesia

*CORRESPONDENCE
Guodong Zhao
✉ m18365806082_1@163.com
Chunkai Wang
✉ chunkaiwang15@163.com

RECEIVED 20 July 2025
REVISED 02 December 2025
ACCEPTED 08 December 2025
PUBLISHED 05 January 2026

CITATION
Xu Z, Zhao G, Wang C and Zhao X (2026) How do village social foundations shape rural residents' participation in digital governance: empirical evidence from village surveys in 11 provinces of China.
Front. Sustain. Food Syst. 9:1669746.
doi: 10.3389/fsufs.2025.1669746

COPYRIGHT
© 2026 Xu, Zhao, Wang and Zhao. This is an open-access article distributed under the terms of the Creative Commons Attribution License (CC BY). The use, distribution or reproduction in other forums is permitted, provided the original author(s) and the copyright owner(s) are credited and that the original publication in this journal is cited, in accordance with accepted academic practice. No use, distribution or reproduction is permitted which does not comply with these terms.

How do village social foundations shape rural residents' participation in digital governance: empirical evidence from village surveys in 11 provinces of China

Zhenzhen Xu[1], Guodong Zhao[2]*, Chunkai Wang[3]* and Xiangyun Zhao[4]

[1]Department of Public Administration, College of Humanities and Social Development, Northwest A&F University, Yangling, China, [2]School of Law, Huazhong University of Science and Technology, Wuhan, China, [3]Department of Sociology and Social Work, College of Humanities and Social Development, Northwest A&F University, Yangling, China, [4]Department of Sociology and Social Work, College of Humanities and Social Development, Northwest A&F University, Yangling, China

Introduction: Digital village initiatives are embedded within village society, and rural Residents' participation in digital governance is shaped by the underlying social foundations of their communities.
Methods: Drawing on state-society interaction theory, this study develops an analytical framework that conceptualizes village social foundations across three dimensions: cultural foundations, organizational foundations, and human-capital foundations. Using survey data from eleven provinces in China, We examine how these foundations influence rural Residents' engagement in digital governance. This issue is especially important in the digital transformation of food systems.
Results and discussion: The results show that stronger village social foundations significantly enhance Residents' participation in digital governance, with organizational foundations exerting the strongest effect. The convenience of digital governance technologies moderates this relationship: higher convenience strengthens the positive impact of village social foundations on participation. Changes in village social foundations also reshape village trust, publicness, and Party-Mass Relations, which in turn alter Residents' levels of digital governance involvement. The influence of village social foundations varies across population groups. Compared with older adults, individuals with lower education, and non-party members, the effects are more pronounced among younger residents, highly educated residents, and Communist Party members. These findings suggest the need to revitalize village cultural resources, strengthen grassroots organizational capacity, and consolidate village-level human capital to further enhance rural Residents' contributions to digital governance and advance the effectiveness of digital village governance.

KEYWORDS

digital governance, food security, village publicness, village social foundation, village trust

1 Introduction

Promoting digital governance in rural areas is not only a strategic choice for rural revitalization but also an urgent response to global sustainable development challenges. According to the Global-Scale Sustainable Development Science Monitoring Report UNESCO (2025): A Decade of Progress from the Perspective of Big Earth Data, released by the

International Research Center of Big Data for Sustainable Development Goals (SDG Center), global progress toward sustainable development has fallen far below expectations since 2015. Eleven out of the eighteen monitored indicators face severe challenges, including core dimensions of human survival and development such as the decline in arable land per capita and the depletion of groundwater resources (UNESCO, 2025). These findings show that improving governance capacity and resilience in rural regions has become a pressing global issue. Despite this urgency, efforts to advance digital governance still confront a marked "digital divide" across rural settings worldwide. In the European Union, digitalization surveys in the agricultural sector show that large farms, farms with better Internet connectivity, and farms receiving professional training exhibit higher adoption rates of digital technologies, while high costs and limited skills remain major barriers (Chinese Academy of Sciences Knowledge Service Platform, 2025). In Indonesia, official statistics indicate that about 3,000 villages still lack any communication signal as of the end of 2025, revealing stark inequalities in digital infrastructure across rural areas globally (ANTARA, 2025). These deficiencies in infrastructure and adoption capacity severely constrain the potential of digital tools to support rural governance.

International research also shows that digital inclusion and digital literacy are essential for enabling rural residents to share the benefits of digitalization. Digital inclusion is considered a key strategy for promoting social integration and addressing population aging (Zhang and He, 2022). Likewise, the effectiveness of e-government initiatives in fostering citizen participation and equalizing public services depends heavily on the digital literacy and skill base of grassroots groups (Isabella et al., 2025).

Rural governance also forms the foundation of China's national governance system. Strengthening digital governance in rural areas is a critical component of advancing national governance modernization. In recent years, both central and local governments have expanded investment in rural informatization. Against this backdrop, the use of digital technologies by rural residents in daily life, administrative services, and public-affairs participation has grown rapidly, moving rural governance into a new stage. The degree of participation in digital governance directly shapes governance performance. Information literacy plays a clear role, as it affects Residents' online abilities and communication networks (Prior et al., 2016). In addition to individual factors, participation takes place within village society and is therefore affected by village-level conditions. Yet current research on rural digital governance participation rarely examines the village perspective. Existing studies focus mainly on digital infrastructure (Zhu and Liu, 2022), democratic awareness, and public consciousness within villages (Zhang et al., 2016). These studies show that higher levels of digital infrastructure raise digital governance participation (Zhao and Sun, 2025), and stronger democratic and public awareness also encourage participation.

Although prior studies provide detailed analyses of participation behavior, they overlook the fact that digital rural governance is embedded within village society and shaped by its social foundations. In practice, some regions exhibit a tendency toward "technological suspension," emphasizing hardware investment and platform construction while overlooking the "social soil" required for technological adoption. Digital governance is not merely a form of tool replacement; it represents a profound reshaping of social relations and governance models. Existing research has failed to address the fundamental influence of village social foundations on digital-village development. In both rural-governance studies and agricultural-sustainability research, social foundations are an essential but underexplored lens. Differences in village social foundations lead to variation in digital-governance participation. Because the information technologies, organizational models, and governance mechanisms applied in digital governance are not endogenous to rural society, tensions may arise between externally introduced digital systems and internal village norms, potentially intensifying governance conflicts (Liu and Lin, 2021). Digital technologies are reshaping entire food systems—from production to consumption. As direct participants and stewards of the food system, rural residents play a decisive role in the success of digital-village development, with implications for national food security and sustainability.

Global experience shows that digital technologies can strengthen the resilience and sustainability of smallholder livelihoods. Research in development economics and agricultural sustainability demonstrates that mobile messaging, mobile applications, and digital platforms providing market prices, input information, and weather services reduce information asymmetries, improve production and marketing decisions, increase income, and enhance farmers' capacity to manage climate risks (Aker and Mbiti, 2010). Randomized controlled trials in Africa and Latin America also show that providing smallholders with price information from major markets and tailored weather forecasts improves their decisions regarding planting, harvesting, and selling (Camacho and Conover, 2019). Digital agriculture, smart farming, and "Agriculture 4.0″ reduce resource waste and environmental pressure through precision fertilization and irrigation (Klerkx et al., 2019). These findings indicate that digital rural governance is not only a key component of national governance modernization but also a critical tool for promoting green transformation in food systems and achieving sustainable development goals. Its effectiveness, however, depends heavily on the social foundations into which technologies are embedded and the actual level of resident participation. Digital rural governance is therefore not only a technical issue but also a social one. It must account for rural Residents' preferences and motivations, mobilize their willingness to participate, and acknowledge the role of village-level cultural, organizational, and human-capital foundations. Only by examining how internal social elements shape participation can we clarify the relationship between technological governance and village society, unlock the endogenous drivers of digital-village development, and strengthen the social foundation of food security.

In this context, this study uses the theory of State-Society interaction to examine the internal logic linking village social foundations and rural Residents' participation in digital governance. Drawing on survey data from 11 provinces in China, we apply an ordered probit model to test the effects and mechanisms of village social foundations. The contributions of this study are threefold. First, while most research focuses on individual factors or public infrastructure, we introduce a village-level perspective centered on foundational social structures. Based on the embedded State-Society framework, we explore how digital governance integrates with rural social systems. Second, previous studies lack a systematic indicator system for village social foundations. We construct such a system across three dimensions—cultural foundations, organizational foundations, and human-capital foundations—to identify how each dimension affects participation. Third, we show that village social

foundations strengthen digital-governance participation through three mechanisms: village trust, village publicness, and Party-Mass Relations. These mechanisms help clarify the internal logic of digital participation among rural residents.

2 Theoretical analysis and research hypotheses

2.1 Digital rural governance in the perspective of state-society interaction

Digital rural governance is both the core task and key objective of digital-village construction. It refers to the use of digital technologies to address intra-village problems and conflicts and to meet the diverse needs of rural residents. As a modern governance tool, "digital" technology carries traits of modernity, rationalization, and technologization that differ from traditional elements within rural society. It is not only a governance resource, but also an important instrument and symbol through which state power and state will extend into the countryside. Thus, the process of digital rural governance is a crucial component of state-building in rural areas and a process through which state power penetrates village society. By exercising foundational power, the state penetrates into society and thereby gains authority over political decision-making in the sphere of social reproduction. Yet state-building ultimately rests on citizens' recognition of the state. Although farmers' participation in digital governance is an individual-level behavior, their level of participation is also an important indicator of the effectiveness of state-building and a concrete manifestation of state will.

It is important to note that the relationship between the state and society is not a zero-sum game. The state is not a supreme authority standing above society and demanding unconditional obedience. In practice, state and society are more often characterized by mutual constraint and cooperation (Mann, 2007). Joel Migdal advances a research orientation of seeing the "state in society" and the "limited state." He argues that the concept of the state includes both the "state in ideas" and the "state in practice." In ideas, the state appears as an independent concept. In practice, however, "the state is only one organization, or a set of organizations, among many social organizations," and it is not independent from society; rather, its interaction with other social forces may even weaken its authority, and its capacity is inherently limited (Migdal, 2013). Peter Evans likewise emphasizes that the state does not stand apart from society. He distinguishes predatory states from developmental states. Predatory states pursue their interests at the expense of society, whereas developmental states are deeply embedded in society. Only by embedding itself in society and engaging with citizens' everyday lives can the state gain community trust and thereby access social resources (Evans, 1997). In this sense, as the state strengthens its presence in society, society in turn reacts upon the state. Through their interaction, the state and society continually reshape one another.

From the perspective of State-Society interaction, state-building must take social factors into account. Within this framework, digital rural governance is not only about the reorganization of village society by state power; it also reshapes the capacity structure of village society itself. In digital-village practice, farmers' participation in digital governance—as a manifestation of state will—must be understood in relation to the social conditions within the village. As a governance instrument of the state, technological governance in rural areas must be aligned with the social foundations of the countryside (Han, 2021). Otherwise, the specific conditions and differentiated needs of rural society will be overlooked, leading to "technological involution" and "digital suspension," in which digital governance becomes a mere formality without real effectiveness (Liu and Luo, 2022).

More specifically, this study defines "village social foundations" as the non-formal and non-economic structural forces within the rural field that support collective action, sustain social integration, and facilitate the implementation of state policies. Based on State-Society interaction theory, digital governance is a formal institutional arrangement promoted by the "state," and its effectiveness depends on the extent to which it is integrated with the existing foundations of "society." Accordingly, we identify cultural foundations, organizational foundations, and human-capital foundations as the core dimensions of village social foundations. Together, these three dimensions constitute the social capital and normative system necessary for digital governance in practice. Cultural, organizational, and human-capital foundations differ from directly deployable economic capital (such as village fiscal resources or individual income) and from top-down political power structures (such as administrative hierarchies). Economic resources are important, but they primarily influence the initial diffusion of digital devices—whether residents can access and use them at all. By contrast, social foundations shape Residents' willingness to participate and their patterns of sustained, in-depth engagement—whether they are willing to use digital tools and whether they can use them well. This is the central concern of the present study.

From the perspective of food-system digitalization, digital rural governance is not a one-way extension of state capacity; it is a process in which state institutions and village social foundations jointly shape outcomes for agricultural sustainability. On the one hand, the state builds digital infrastructure and creates administrative and service platforms that incorporate price information, subsidy policies, technical standards, and climate information into an integrated institutional supply system. On the other hand, only when these digital tools are embedded in local social networks and aligned with existing cultural norms, cooperative organizations, and rural elites can smallholders effectively use digital platforms to obtain inputs and financial services, receive climate and disaster warnings, and access more transparent information on market prices and demand, thereby improving production and marketing decisions (Fabregas et al., 2019). Existing research shows that the effectiveness of digital agriculture depends largely on local social capital, organizational resources, and human capital, rather than solely on the sophistication of the technology itself (Toyama, 2015).

Therefore, understanding digital rural governance from a State-Society interaction perspective requires treating digital governance as the outcome of alignment between state institutional design and village social foundations. The state extends its foundational power through digital means, while village social foundations—through networks of trust, structures of publicness, and Party-Mass Relations—shape farmers' responses to digital governance and their depth of participation. Together, these factors determine the real effectiveness of digital governance in promoting food security and rural sustainability.

2.2 The impact of village social foundations on rural residents' digital governance participation

In the process of "digital technologies entering the countryside," technology is not omnipotent. Digital-village construction is not merely a matter of technological innovation or external driving forces. It is not enough simply to introduce and diffuse technologies into villages. More importantly, these technologies must adapt to the village social environment, be accepted and used by rural residents, and generate real governance effects. This requires close attention to existing institutional, cultural, and organizational elements within village society. For a long time, Chinese rural society has been organized as a "society of acquaintances" and a rural community grounded in kinship and local ties. On this basis, stable and mature patterns and logics of governance have formed. Cutting-edge digital technologies do not naturally fit with these local structures and may even generate tensions (Liu and Lu, 2022). From the perspective of economic sociology, the key issue in digital rural governance is embeddedness: digital technologies must be integrated into village social foundations and linked to specific participation contexts in village governance, in order to activate local elements and mobilize the endogenous motivation of rural residents (Qiu et al., 2022). If digital technologies cannot be effectively aligned with local social structures, they will not produce substantive effects, no matter how convenient they appear.

International experience confirms this conclusion. In developing regions and in countries with marked urban–rural differences, digital literacy, digital inclusion, and citizens' trust in e-government jointly shape grassroots Residents' ability to participate in public affairs online and to use digital public services (Bai and Yang, 2025). Village social foundations influence how digital technologies are used and promoted. Under different social foundations, digital technologies must adjust their modes of entry into villages, their functional design, and their role in governance systems to fit village specificities and diverse resident needs. Strong village social foundations can facilitate the localization of digital tools, enhance rural Residents' participation in digital governance, and help to avoid the problem of "digital suspension." Based on the above discussion, we propose the following hypothesis:

> *Hypothesis* 1: Digital-village construction is embedded in village social foundations, and stronger village social foundations increase rural Residents' participation in digital governance.

With respect to the mechanisms through which village social foundations influence participation, we argue, drawing on existing literature and theoretical logic, that village social foundations affect digital governance participation through three channels: strengthening village trust, enhancing village publicness, and deepening Party-Mass Relations.

First, village trust. Trust is the basis of individual action and social order, and it is a precondition for rural residents to participate in village governance. Village trust refers to a community order in which individuals, embedded in village systems through social norms and community rules closely linked to local culture, form positive expectations of one another within the framework of the village community (Hu, 2004). Village trust is thus a key dimension of rural governance. Only in a context of mutual trust can village collectives gain mobilizing capacity and can villagers be willing to participate in governance affairs (Hu and Huang, 2022). In digital-village construction, village trust helps overcome collective-action dilemmas, increases villagers' recognition of digital governance, and encourages their participation in digital initiatives. At the same time, village trust is grounded in specific social foundations and is shaped by institutional factors within the village (Wang, 2007). Cultural, organizational, and human-capital foundations are central components of village social foundations and form the preconditions for village trust. Strong cultures of filial piety, mutual assistance, and local identity increase interaction and mutual dependence among villagers. Through long-term, repeated interactions, village trust is gradually built. Well-structured village committees and farmer organizations with strong capacity for action provide a solid organizational foundation, help to mediate conflicts, meet Residents' needs, and strengthen villagers' organizational identification, thereby fostering village trust. Based on this logic, we propose the following hypothesis:

> *Hypothesis 2*: Village social foundations increase rural Residents' participation in digital governance by strengthening village trust.

Second, village publicness. Village publicness is the basis for maintaining social order in villages. It arises from clan-based mutual help, collective cooperation, and formal institutional arrangements that satisfy shared needs (Li, 2020). It links governance actors to the provision of public goods and implies a shift from private or family-based particularistic interests to the pursuit of collective interests at the village level (Feng, 2021). Village publicness mobilizes public participation, nurtures public spirit, and supports collective action by expanding public space. The stronger the village publicness, the more actively rural residents participate in governance affairs. In digital-village construction, the diffusion of wireless networks, information platforms, and social-media tools reshapes interpersonal interaction, the content of public affairs, and the direction of village development. Access to richer and more diverse information further enhances villagers' participation awareness and managers' sense of responsibility at the level of publicness (Ma and Zhang, 2020). Thus, village publicness can enhance villagers' agency and enthusiasm, and help avoid digital suspension. Village publicness is not innate; it is cultivated within village social foundations. Deep cultural roots strengthen villagers' sense of place, cultivate public spirit, and shape a collective orientation toward action. Solid social foundations provide the conditions for villagers to care about village development and transformation. Through the influence of cultural, organizational, and human-capital foundations, villagers' sense of responsibility and mission is reinforced, making them more willing to embrace change and participate in digital governance. Accordingly, we propose the following hypothesis:

> H*ypothesis* 3: Village social foundations increase rural Residents' participation in digital governance by enhancing village publicness.

Third, Party-Mass Relations. Party-Mass Relations are a central way in which the Communist Party of China participates in social governance and they shape the basic logic of rural governance. Farmers live in concrete environments and have rich emotional and

practical needs. Rural governance must respond to these specific people, and Party-Mass Relations serve as a key pathway for doing so, enabling governance to address farmers' real concerns (Du, 2020). In digital-village construction, close Party-Mass Relations not only strengthen villagers' trust in party organizations, but also facilitate the use of party organizations to publicize digital technologies and promote digital initiatives. Villagers can transfer their trust in party organizations and party members to digital-village activities and thus participate more actively in digital governance. Strong Party-Mass Relations are inseparable from village social foundations. Village culture functions as a natural adhesive for Party-Mass Relations. Village regulations, customs, and other cultural practices enhance village cohesion and help to maintain close ties between the Party and the masses. Cultural, organizational, and human-capital foundations provide the basis for close Party-Mass Relations. In the process of digital technologies entering the countryside, they promote the acceptance and use of digital tools among different groups, enhance the capacity and willingness of diverse actors to participate in governance, and foster the formation of a governance community (Su and Peng, 2022). Based on this reasoning, we propose the following hypothesis:

> *Hypothesis* 4: Village social foundations increase rural Residents' participation in digital governance by strengthening Party-Mass Relations.

In addition, within villages, the impact of village social foundations on Residents' participation in digital governance is heterogeneous across different groups. As digitalization advances rapidly, middle-aged and older adults lag far behind other groups in access to and use of information technologies due to constraints related to technology, institutions, culture, and their own capacities. They may become "digital refugees," excluded from the digital society (Lu and Wei, 2021). Younger residents are thus more likely to participate in digital-village activities. Because digital tools require specific skills and usage rules, residents with higher educational attainment tend to have stronger digital literacy and greater capacity for the effective use of digital technologies. Compared with those with lower education, highly educated rural residents are more motivated to participate in digital governance. At the same time, in the process of "digital technologies entering the countryside," the diffusion and application of digital governance tools often proceed in a stepwise fashion. Party members typically play a more prominent role in public-affairs participation and collective activities. As a result, they also tend to be more active in digital-village governance. Based on this, we propose the following hypothesis:

> *Hypothesis* 5: The positive impact of village social foundations on digital governance participation is more pronounced among younger residents, residents with higher education, and party members.

Building on the above literature review and research hypotheses, we construct an analytical framework to capture how village social foundations influence rural Residents' participation in social governance. The framework is presented in Figure 1.

3 Data source, variable description, and model specification

3.1 Data source

The data used in this study originate from a questionnaire survey conducted by the research team in August 2022 across 11 provinces in mainland China, including Jiangsu, Zhejiang, Guangdong, Shandong, Fujian, Henan, Shaanxi, Gansu, Sichuan, Ningxia, and Yunnan. The survey covered 41 counties (districts) and 205 villages. A multistage stratified sampling method was employed.

First, based on economic development levels and social governance conditions, five provinces in eastern China (Jiangsu, Zhejiang, Guangdong, Shandong, Fujian) and six provinces in central and western China (Henan, Ningxia, Shaanxi, Gansu, Sichuan, Yunnan) were selected as primary sampling units. Second, 2–4 counties or districts were selected from each province, including one jurisdiction under the provincial capital and additional jurisdictions from other prefectures. Finally, 3–5 rural communities were chosen within each county/district, and 20–30 households were randomly sampled in each village according to population size and distribution. In each household, one member familiar with household and village conditions was selected for face-to-face interviews.

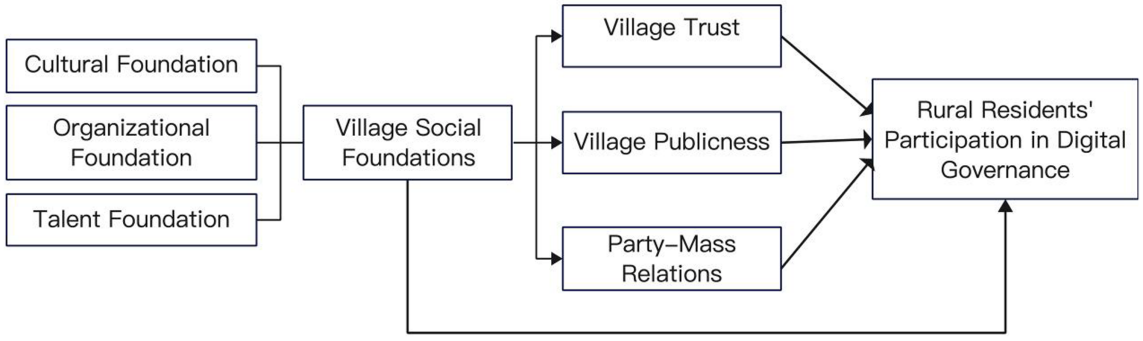


FIGURE 1
The impact and mechanisms of village social foundations on rural residents' participation in digital governance.

The questionnaire included modules on household characteristics, village conditions, digital governance, and individual cognition. A total of 2,300 questionnaires were distributed, and 2,290 were returned, yielding a response rate of 99.6%. After conducting validity checks and removing questionnaires with excessive missing values, abnormal responses, or inconsistencies, 2,279 valid observations of rural residents were retained for analysis.

3.2 Variable description and descriptive statistics

3.2.1 Dependent variable: digital governance participation

Digital governance participation is measured from three dimensions: digital life applications, digital government services, and participation in digitalized village public affairs. Digital life applications were measured by the question: "Have you used mobile phones or self-service online platforms to handle daily affairs (e.g., payment of social security fees, electricity bills, ID information registration)?" Responses were coded as 1 for "yes," 0 for "no," and "uncertain" was treated as missing. Digital government services were measured by the question: "Have you used grassroots government platforms to handle administrative services online (e.g., household registration, online ID card processing)?" Coding followed the same rules as above. Participation in digitalized village public affairs was measured by the question: "Have you posted messages in the village WeChat group or read village affairs information on the smart governance platform?" Responses were coded as 1 for "yes" and 0 for "no."An aggregate index was constructed by summing the three items to assess overall participation in rural digital governance.

3.2.2 Core explanatory variable: village social foundation

There is no unified standard for measuring the village social foundation in existing research. Studies generally examine social relations, rural lifestyles (Li and Song, 2022), organizational structures (Zhao et al., 2022), cultural activities, and spatial environments (Wang, 2022), encompassing both material and non-material elements of village society. However, such broad perspectives risk conceptual overstretch. A more fundamental analytical approach is required. From the perspective of economic sociology, cultural elements such as values and cognition are central determinants of action—reflected in Weber's analysis of Protestant ethics and Bourdieu's concept of habitus. Organizational sociology emphasizes that organizations, as intentionally constructed social units, shape collective behavior and social outcomes. Meanwhile, elite theory highlights the important influence of local leaders and elites on community development trajectories. Drawing on these perspectives and considering the specific characteristics of rural society, this study conceptualizes the village social foundation through three dimensions: village cultural foundation, village organizational foundation, and village talent foundation. Based on these dimensions, an index system was developed and nine initial indicators were designed. Exploratory factor analysis (EFA) was applied to refine the measurement. Six indicators were retained after EFA (see Table 1). The KMO statistic for the six-item scale is 0.55—slightly below the common threshold of 0.6—but Bartlett's test is significant at 0.000 (<0.05), indicating sufficient inter-item correlations for factor analysis. Using principal component analysis, three common factors with eigenvalues greater than 1 were extracted. Following rotation and interpretation based on the loadings, the three factors were named village cultural foundation, village organizational foundation, and village talent foundation, with a cumulative variance contribution of 62.27%. The overall village social foundation index is computed as a weighted composite of the three factor scores, where each factor is weighted by its proportion of variance contribution. All retained items have factor loadings above 0.50, demonstrating adequate convergent validity.

TABLE 1 Indicators and descriptions of the village social foundation.

Dimension	Specific measurement indicators	Factor loading
Village cultural foundation	Whether neighbors can borrow money from each other (culture of mutual help)	0.78
	Whether villagers feel proud of being a member of the village (local identity culture)	0.80
Village organizational foundation	Whether village cadres possess high authority/ prestige	0.59
	Whether villagers participate in consultation on village public affairs	0.87
Village talent foundation	Whether the village has capable mediators	0.76
	Whether the village has capable party-building talents	0.75

3.2.3 Control variables

A large body of literature has confirmed that rural Residents' individual characteristics, household attributes, village conditions, and regional environments all have significant effects on their participation in digital rural governance (Su and Peng, 2021). Drawing on existing studies, this paper includes four groups of control variables: Individual characteristics and cognitive variables, including gender, age, education level, ethnicity, political identity, and whether the respondent works outside the village. Household characteristics, measured by annual household income. Village characteristics, including whether communication networks are convenient and whether the village has WeChat mini-programs for governance services. Regional characteristics, distinguishing whether the respondent lives in a developed eastern region or a less developed central–western region.

3.2.4 Instrumental variables

Although controlling for observable omitted variables can partly mitigate endogeneity, unobservable factors may still bias the estimates. In addition, a bidirectional causal relationship may exist between village social foundations and rural Residents' digital governance participation. Therefore, this study employs an instrumental variable approach to address endogeneity. We select the mean value of social foundation indicators at the district (county) level as the instrumental

variable. This variable affects individual digital governance participation mainly through "peer effects" or "social multiplier effects," which represent legitimate channels through which the broader social environment shapes individual behavior. The district-level mean forms the macro-context of the village's social environment. It influences regional trust norms, behavioral expectations, and information diffusion, thereby indirectly shaping individual decisions. In the empirical model, we directly control for the respondent's own village-level social foundation. This treatment isolates the indirect pathway of the instrument and avoids direct effects on individual outcomes.

Villages within the same district tend to share similar social foundations. Meanwhile, the district-level mean of digital technology usage has no direct influence on the digital governance participation of individuals in a given village. Thus, the instrumental variable meets both relevance and exogeneity conditions.

3.2.5 Mediating variables

Three mediating variables are included: village trust, village publicness, and party-mass relations.

Village trust is measured using the question: "How much do you trust people in your hometown?" with values ranging from 1 ("very distrustful") to 5 ("very trustworthy"). Village publicness is measured by villagers' perceived level of participation in public affairs: "How would you rate the level of villager participation in your village's governance?" with values from 1 ("very inactive") to 5 ("very active"). Party-Mass Relations are measured by respondents' satisfaction with the relationship between Party members and ordinary villagers: "How satisfied are you with the relationship between Party members and the masses around you?" with values from 1 ("very dissatisfied") to 5 ("very satisfied").

3.2.6 Moderating variable

The moderating variable is the convenience of digital governance technology in the village. It includes two aspects: Network accessibility, measured by whether respondents agree that Wi-Fi can be accessed conveniently in the village ("yes" = 1; "no" = 0). Ease of operating digital governance platforms, measured by agreement with the statement: "Using mobile phones in village governance is very convenient." ("yes" = 1; "no" = 0). These two items are summed to construct an overall indicator of the convenience of digital governance technology.

Descriptive statistics for all variables are presented in Table 2.

3.3 Model specification

This study examines the impact of village social foundations on rural Residents' participation in digital governance. Since participation in digital governance is an ordinal variable, we employ an ordered probit model for estimation. The model is specified as follows:

$$Prob\left(digital_i = W|X_i\right) = \alpha_0 + \alpha_1 social_i + \alpha_2 control_i + \sigma_i \quad (1)$$

TABLE 2 Variable definitions, coding, and descriptive statistics.

Variable	Coding description	N	Mean	SD	Min	Max
Digital governance participation (score)	Sum of the three binary items	1963	1.710	1.124	0	3
Village social foundation	Sum of indicator items	2,279	0.731	0.246	0	1.152
Gender	Male = 1, Female = 0	2,279	0.520	0.500	0	1
Age	Respondent's actual age	2,279	42.522	17.135	18	90
Ethnicity	Han = 1, Minority = 0	2,279	0.927	0.260	0	1
Education level	Primary or below = 1; Middle school = 2; High school = 3; College = 4; Graduate = 5	2,279	2.567	1.125	1	5
Political identity	Party member = 1, Non-member = 0	2,279	0.122	0.327	0	1
Migrant work experience	Yes = 1, No = 0	2,279	0.595	0.491	0	1
Household income	Income groups: <20 k = 1; 20 k–50 k = 2; 50 k–100 k = 3; 100 k–150 k = 4; 150 k–200 k = 5; 200 k–500 k = 6; >500 k = 7	2,279	2.883	1.307	1	7
Village network quality	Good = 1, Poor = 0	2,279	0.670	0.470	0	1
Village mini-program availability	Governance mini-program available = 1, Not available = 0	2,279	0.338	0.473	0	1
Region	Eastern region = 1, Central/Western region = 0	2,279	0.254	0.435	0	1
Convenience of digital technology	Sum of two binary items	2,279	0.872	0.693	0	2

TABLE 3 Effects of village social foundations on rural residents' participation in digital governance.

Variables	Participation in digital governance			
	(1)	(2)	(3)	(4)
Village social foundation	0.816*** (0.15)			
Village cultural foundation		0.446 (0.363)		
Village organizational foundation			1.952*** (0.203)	
Village talent foundation				0.158 (0.232)
Gender	0.09 (0.06)	0.086 (0.061)	0.05 (0.061)	0.087 (0.06)
Age	−0.02*** (0.003)	−0.018*** (0.003)	−0.021*** (0.003)	−0.018*** (0.003)
Ethnicity	0.219 (0.208)	0.203 (0.201)	0.289 (0.203)	0.2 (0.2)
Education level	0.054 (0.038)	0.063* (0.037)	0.073* (0.038)	0.059 (0.038)
Political identity	0.149 (0.095)	0.266*** (0.091)	0.087 (0.092)	0.258*** (0.093)
Household income	0.034 (0.038)	0.049 (0.04)	0.038 (0.039)	0.048 (0.04)
Migrant work experience	0.172** (0.073)	0.173** (0.072)	0.165** (0.071)	0.17** (0.072)
Village network signal quality	0.132 (0.083)	0.136* (0.082)	0.167** (0.077)	0.135* (0.081)
Village governance mini-app	0.904*** (0.092)	0.938*** (0.092)	0.85*** (0.089)	0.931*** (0.093)
Region (East = 1)	−0.314** (0.122)	−0.344*** (0.124)	−0.28** (0.109)	−0.34*** (0.125)
Observations	1963	1963	1963	1963
R^2	0.129	0.118	0.139	0.118

$p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$; Cluster-robust standard errors (clustered at the village level) are reported in parentheses.

Where, $digital_i$ is the ordered dependent variable, taking values W = 0, 1, 2, or 3. $social_i$ represents the village social foundation for respondent i, and control_i includes control variables such as gender, age, education level, political identity, household income, and village network conditions. α_0, α_1 and á$_2$ are model coefficients, and σ_i is the random disturbance term. Because respondents from the same village may share similar characteristics, we cluster the standard errors at the village level using cluster-robust standard errors.

4 Empirical results and analysis

4.1 Common method bias test

Using a single measurement method in questionnaire surveys may generate systematic bias and lead to common method bias. This study adopts four strategies to address this concern. First, the core variables were measured using different numbers of alternative responses and varied wording. Respondents also completed the questionnaire anonymously to reduce response bias caused by subjective anxiety. Second, the survey relies on a large sample covering multiple provinces. This helps reduce data bias arising from differences in individual characteristics. Third, the questionnaire items did not use the actual names of the variables. This avoids inaccurate answers caused by respondents' preconceived expectations. Fourth, Harman's single-factor test was employed to assess the magnitude of common method bias. The results show that four factors had eigenvalues greater than 1, and the first factor explained 18.5% of the variance (below the 40% threshold). This indicates that common method bias is well controlled in this study.

TABLE 4 Marginal effects of the ordered probit model at sample means.

Variables	Participation in digital governance			
	Y = 0	Y = 1	Y = 2	Y = 3
Village social foundation	−0.202*** (0.038)	−0.105*** (0.023)	0.032** (0.014)	0.275*** (0.053)
Control variables	Controlled	Controlled	Controlled	Controlled
Observations	1963	1963	1963	1963
R^2	0.129	0.118	0.139	0.118

$p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$; Cluster-robust standard errors (clustered at the village level) are reported in parentheses.

4.2 Baseline regression results

4.2.1 The impact of village social foundations on rural residents' participation in digital governance

Table 3 reports the effects of village social foundations on rural Residents' participation in digital governance. Column (1) presents the impact of overall village social foundations. Columns (2)–(4) show the effects of the three sub-dimensions—cultural foundations, organizational foundations, and talent foundations. Overall, village social foundations significantly increase rural Residents' digital governance participation, with results significant at the 1% level. Digital technologies are embedded in village social structures. When village social foundations are stronger, the alignment between "digital initiatives entering villages" and local needs improves. This helps overcome digital "suspension" and increases Residents' willingness to participate in digital governance (Table 4).

We further report the marginal effects at the sample mean. These values show how a one-unit change in the key explanatory variable alters the probability of each participation category when continuous variables are set at their sample means and categorical variables at their base categories. The marginal effects reveal a clear pattern: stronger village social foundations reduce the probability of belonging to lower participation categories and increase the probability of belonging to higher participation categories. Specifically, at the average individual level, a one-unit increase in village social foundations reduces the probability of falling into the "no participation" and "low participation" groups by 20.2 and 10.5%, respectively. In contrast, the probability of falling into the "high participation" and "very high participation" groups increases by 3.2 and 27.5%, respectively. These marginal effects provide stronger evidence for the positive role of village social foundations.

Regarding the three sub-dimensions, only organizational foundations significantly promote digital governance participation. Cultural foundations and talent foundations do not show significant results. One main reason is that digital rural construction is still in a development stage. Its implementation depends on grassroots organizational capacity. Excessive attention to advanced technologies often leads to neglect of the institutional and organizational structures needed to support them (Shen and Du, 2022) As a result, it becomes difficult to enhance Residents' participation in digital governance. Cultural norms such as trust and reciprocity are important, yet without organizational channels for expression, they remain informal and personal in nature. Grassroots organizations transform these cultural norms into concrete rules, institutions, and public activities—for example, points-based systems, consultation meetings, and collective events. These mechanisms provide stable platforms for the formation and operation of trust and public-mindedness. For instance, Taiping Village in Pidu District, Chengdu, Sichuan Province, shifted from traditional paper-based disclosure to a fully digital information system. Core modules such as public affairs disclosure, major event records, and announcements were moved online. Residents now receive real-time updates on collective asset management, road maintenance budgets, environmental improvement progress, and policy information. This eliminates time and space constraints. During the early promotion stage, village cadres and volunteer teams mobilized households through door-to-door visits. They combined village meetings, party member meetings, Residents' representative meetings, and cultural activities such as food festivals and film screenings to guide villagers to register on the platform. Digital services including "water-fee reservations," "certificate issuance," "preferential card applications," and "elderly certification" were integrated into the "service hall" module. These efforts helped residents form the habit of "turning to the platform when needed," and more than 90% of villagers eventually joined the digital platform. Therefore, stronger organizational foundations enhance a village's capacity to promote digital technologies, improve Residents' familiarity with and acceptance of digital governance, and strengthen their sense of participation and benefit.

Compared with Liang Shuming's earlier concern that "written language could not reach the countryside," which caused cultural mismatches and limited farmers' active participation, the phenomenon of "digital technologies not entering the countryside" is far more complex (Liang, 2011). Digital technologies require villagers who understand and can operate digital devices and platforms (Liu, 2022). Participation also depends on demonstration effects and social learning. In villages with stronger cultural foundations, norms of mutual assistance and communication promote interaction and cohesion among villagers. This, in turn, enhances digital governance participation across different groups.

4.2.2 Moderating effect of the convenience of digital governance technology application in villages

Due to regional differences in the extent of digital governance application, rural Residents' acceptance of digital governance technologies varies across locations, which in turn affects the influence of village social foundations on their participation in digital governance. To examine the moderating effect of the convenience of digital governance technology application in villages, this study adds an interaction term between village social foundations and the convenience of digital governance technologies ($social_i \times convenience_i$) to the baseline regression model (1), and expands the equation as follows:

$$Prob\left(digital_i = W|X_i\right) = \beta_0 + \beta_1 social_i + \beta_2 social_i \times convenience_i + \beta_3 control_i + \varepsilon_i \quad (2)$$

In Equation 2, convenience is an ordinal variable that takes values from 0 to 3, with larger values indicating greater convenience in the application of digital governance technologies. The interaction coefficient β_3 captures the moderating effect of technological convenience on the relationship between village social foundations and rural Residents' participation in digital governance. The estimation approach remains consistent with that of Equation 1.

Table 5 shows that the interaction term between village social foundations and the convenience of digital governance technologies is positive and statistically significant. This indicates that the effect of village social foundations on Residents' participation in digital governance is moderated by the convenience of digital governance technology application. When digital governance technologies in a village are more convenient to use, the positive impact of village social foundations on Residents' participation is stronger; conversely, when such technologies are less convenient, the effect becomes weaker. This finding helps explain why Residents' participation in social governance may differ across areas with similar village social foundations.

TABLE 5 Test of the moderating effect of digital governance technology convenience.

Variables	(1)
	Digital governance participation
Village social foundation	0.4** (0.202)
Convenience of digital governance technology	−0.194 (0.15)
Village social foundation × Technology convenience	0.495*** (0.186)
Control variables	Controlled
Regional fixed effects	Controlled
Observations	1963
R^2	0.134

Specifically, besides paying attention to village social foundations, it is also necessary to recognize that the acceptance and ease of use of digital technologies are closely tied to their actual application. Under the premise of strengthening village social foundations, further improving the convenience of accessing and using digital technologies is crucial for enhancing participation in village-level digital governance.

To further examine how the moderating variable—the convenience level of digital governance technology in the village—affects the relationship between village social foundations and rural Residents' participation in digital governance, we calculate the marginal effects at each level of technology convenience. As shown in Figure 2, the results indicate that the convenience of digital governance technology exerts a significant moderating effect on the relationship between village social foundations and Residents' participation in digital governance. As the convenience level of village digital governance technology increases from Level 0 to Level 2, the positive effect of village social foundations on Residents' participation shows a clear upward trend. Specifically, when technology convenience is at the lower Level 0, the promoting effect of village social foundations on participation is relatively limited; when convenience increases to Level 1, the positive effect of social foundations begins to strengthen significantly; and under the highest convenience condition (Level 2), the promoting effect of village social foundations on Residents' participation in digital governance reaches its maximum. This result suggests that a more complete digital technology infrastructure can effectively amplify the positive influence of village social foundations, creating more favorable conditions for residents to engage in digital governance.

4.3 Robustness checks

4.3.1 Endogeneity test

Since the endogenous variable—village social foundations—is an ordered variable, and the Ivoprobit model can only address cases where the endogenous regressor is continuous, this study employs the Conditional Mixed Process (CMP) estimation method. The Ivoprobit_CMP approach places fewer restrictions on endogenous explanatory variables and allows both continuous and non-continuous forms. Therefore, the Ivoprobit_CMP estimator is used to re-estimate the model to further verify the robustness of the results. As shown in Table 6, we conduct weak-instrument tests and endogeneity tests to assess the validity of the instrumental variable. First, the first-stage regression results show that the county-level mean of social foundations is significantly correlated with village social foundations at the 1% statistical level, indicating that the instrumental variable satisfies the relevance requirement. The first-stage F-statistic is 70.41, demonstrating that the selected instrumental variable is not weak and meets the basic criteria for instrument validity. According to the estimation results, the coefficient of the core explanatory variable remains stable, indicating that after addressing endogeneity, village social foundations still exert a significant positive effect on rural Residents' participation in digital governance.

4.3.2 Robustness check I: generalized ordered probit model

Since the ordered probit model relies on the parallel regression assumption, this study first conducts a Brant test. The test result shows a p-value of 0.000, which is lower than 0.05, indicating a possible violation of the parallel regression assumption. Therefore, this study further employs the Generalized Ordered Probit (GOP) model as a

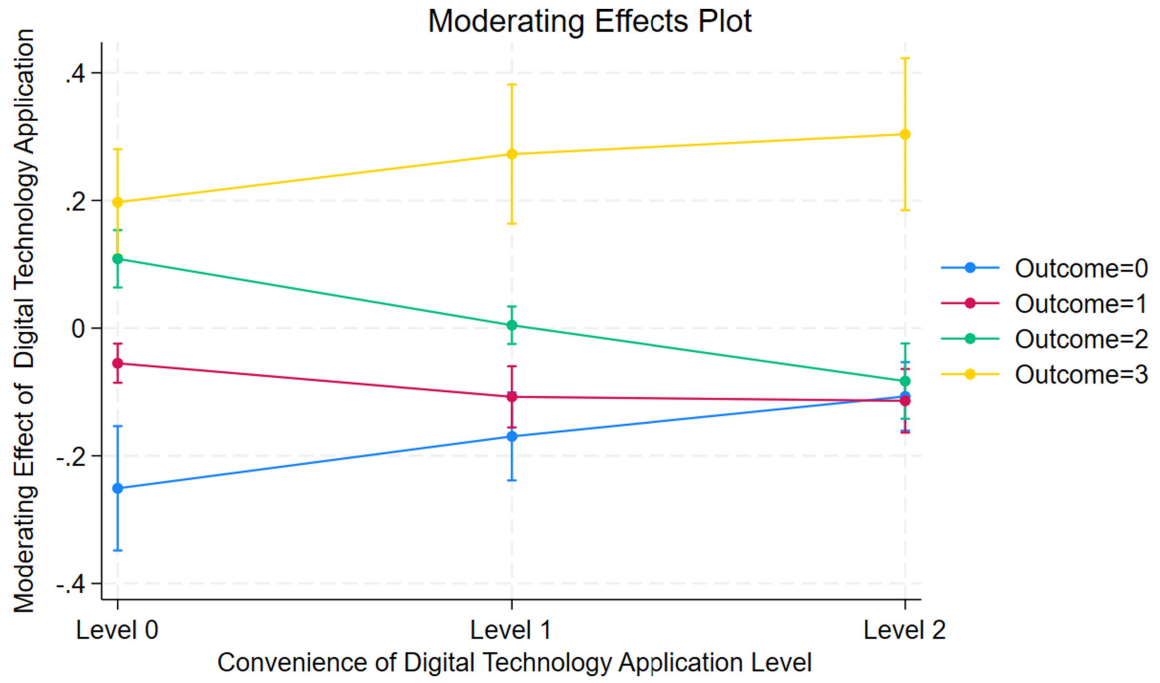


FIGURE 2
Conditional marginal effects of digital governance technology convenience in villages.

TABLE 6 Instrumental variable test results.

Model	Ivoprobit_CMP	
	First stage	Second stage
Village social foundations	—	0.472* (0.246)
County-level mean of social foundations (instrument)	0.921*** (0.040)	—
Control variables	Controlled	Controlled
Observations	2,279	2,279
Cragg-Donald Wald *F*	70.41	

robustness check. The GOP model relaxes the parallel regression constraint and allows the coefficients of the same explanatory variable to vary across different threshold levels, making it more consistent with empirical reality than a standard ordered probit model.

The estimation results are presented in Table 7. The effects of the explanatory variables differ significantly across the cumulative probability equations. Specifically, village social foundations show a significantly positive effect in the first cumulative probability equation ($\beta = 0.472, p < 0.01$), indicating that stronger village social foundations increase the probability that rural residents fall into a higher category of digital governance participation rather than the lowest category. In subsequent equations, the coefficient of village social foundations becomes even more significant, demonstrating that its influence grows stronger for higher levels of digital governance participation.

4.3.3 Robustness check II: the effect of social foundations on participation in single domains of digital governance

To further verify the robustness of the baseline results—where the dependent variable is the overall level of rural Residents' participation in digital governance—this section examines whether village social foundations also affect participation in specific digital governance domains. Table 8 Columns (1)–(3) reports the estimation results for digital participation in daily life services, government affairs, and village public affairs, respectively. The findings indicate that strong village social foundations significantly promote Residents' participation in all three domains: digital life governance, digital government service use, and digital public affairs engagement. This suggests that social foundations exert a consistent and significant influence on rural Residents' digital governance participation, thereby confirming the robustness of the baseline results.

4.4 Mechanism analysis

This study employs a mediation-effect model to examine the underlying mechanisms. The testing procedure is as follows: first, each mediating variable—village trust, village publicness, and Party-Mass Relations—is separately treated as the dependent variable, while village social foundations serve as the independent variable. Second, both village social foundations and the mediating variable are included in the regression model to assess their effects on Residents' participation in digital governance. Let M denote the mediating variable; if both coefficients α_1 and β_1 are statistically significant, mediation is present.

$$M_i = \alpha_0 + \alpha_1 social_i + \beta\alpha_2 X_i + \lambda_t + \varphi_i + \mu_i$$

$$\text{digital}_i = \beta_0 + \beta_1 M_i + \beta_2 social_i + \beta_3 X_i + \varphi_i + \mu_i$$

TABLE 7 Robustness check: results from the generalized ordered probit model.

Variables	(1)	(2)	(3)
	P(Y > 0)	P(Y > 1)	P(Y > 2)
Village social foundations	0.472*** (0.176)	0.721*** (0.174)	1.163*** (0.194)
Gender	0.074 (0.083)	0.046 (0.065)	0.134* (0.071)
Age	−0.024*** (0.004)	−0.017*** (0.003)	−0.017*** (0.003)
Ethnicity	0.347 (0.258)	0.109 (0.188)	0.248 (0.226)
Education level	0.098** (0.048)	0.063 (0.039)	0.014 (0.048)
Political identity	0.349*** (0.129)	0.149 (0.111)	0.022 (0.105)
Household income	0.024 (0.035)	0.03 (0.047)	0.045 (0.039)
Migrant work experience	0.213** (0.083)	0.177** (0.079)	0.109 (0.091)
Village network signal	0.171 (0.109)	0.077 (0.085)	0.086 (0.116)
Village governance mini program	0.636*** (0.114)	0.666*** (0.099)	1.131*** (0.098)
Region	−0.424** (0.181)	−0.356*** (0.127)	−0.218 (0.136)
Observations	1963	1963	1963

TABLE 8 Robustness check II: the effect of village social foundations on participation in single domains of rural digital governance.

Variables	(1)	(2)	(3)
	Digital life participation	Digital government service participation	Digital public affairs participation
Village social foundations	0.636*** (0.175)	0.803*** (0.194)	0.611*** (0.171)
Control variables	Controlled	Controlled	Controlled
Observations	2097	2019	2,279
R^2	0.158	0.171	0.15

4.4.1 Village trust mechanism

Columns (1)–(2) in Table 9 report the results. Column (1) shows that village social foundations have a significant positive effect on village trust: stronger social foundations—particularly mutual assistance and neighborhood watch culture—lead to higher levels of interpersonal trust. Column (2) indicates that village trust significantly increases Residents' participation in digital governance. Higher levels of trust strengthen interpersonal interaction and communication, help overcome collective-action constraints in rural communities, enhance

TABLE 9 Mechanism tests: effects of village social foundations on rural digital governance participation.

Variables	Village trust effect		Village publicness effect		Party–mass linkage effect	
	M	Social	M	Social	M	Social
	(1)	(2)	(3)	(4)	(5)	(6)
Village social foundations	1.321*** (0.149)	0.745*** (0.152)	1.918*** (0.129)	0.691*** (0.151)	2.463*** (0.159)	0.672*** (0.157)
Social capital		0.082* (0.045)				
Village publicness				0.09** (0.039)		
Party-mass relations						0.089** (0.044)
Control variables	Controlled	Controlled	Controlled	Controlled	Controlled	Controlled
Sample size	2,279	1963	2,279	1963	2,279	1963
R^2	0.04	0.13	0.085	0.13	0.122	0.13

TABLE 10 Bootstrap test of mediation effects.

Mediation mechanisms linking village social foundations to residents' digital governance participation	Effect	SE	Bias-corrected 95% CI	
			Lower	Upper
Village trust, village publicness, and party-mass relations → residents' digital governance participation	0.246	0.069	0.111	0.382

Residents' identification with digital governance, and ultimately encourage higher participation.

4.4.2 Village publicness mechanism

Columns (3)–(4) in Table 9 present the results. Column (3) shows that village social foundations significantly enhance village publicness. When social foundations are strong, civic spirit tends to be higher, and residents are more willing to participate in village activities. This expands opportunities for interaction among different groups, strengthens social ties, and fosters integration across social strata. Column (4) shows that village publicness significantly promotes digital governance participation. Stronger publicness improves collective-action capacity, reinforces civic awareness and responsibility, and supports higher levels of resident participation in digital governance.

4.4.3 Party-mass relations mechanism

Columns (5)–(6) in Table 9 report the results. Column (5) shows a significant positive relationship between village social foundations and Party-Mass Relations, primarily because strong organizational and talent foundations strengthen community identity and enhance Party–mass cooperation. Column (6) indicates that closer Party-Mass Relations significantly increase Residents' participation in digital governance. Strong Party-Mass Relations reinforce organizational trust and improve the capacity of Party members to guide digital governance, thereby helping to build a multi-actor digital governance community (Shen and Bi, 2021).

To further verify the mediation mechanism, we employ the Bootstrap method (1,000 replications) to test the significance of mediation effects in the non-linear oprobit model. As shown in Table 10, the mediation effects of village trust, village publicness, and Party-Mass Relations are all statistically significant. The 95% confidence intervals do not include zero (LLCI = 0.111, ULCI = 0.382), providing additional evidence that these mechanisms indeed mediate the relationship between village social foundations and Residents' digital governance participation.

4.5 Heterogeneity analysis

Village social foundations exert a significant overall influence on rural residents' participation in digital governance. However, due to individual differences, residents with similar village social foundations may still display different levels of participation. As shown in Table 11, this section examines the heterogeneous effects of village social foundations across different types of residents, focusing on age groups, education levels, and political identity. Age is divided into younger and middle-aged/older groups using 40 years old as the cutoff. Education level is divided into high-education and low-education groups using high school as the threshold. Political identity is divided into Party members and non-Party members.

4.5.1 Residents of different age groups

The regression results show that the effect of village social foundations on digital governance participation is stronger among younger residents than among middle-aged and older residents. This difference stems from the clear generational digital divide in the adoption and use of digital technologies. Younger residents tend to adopt new media more readily, allowing them to use digital tools more skillfully, expand their social networks, and engage in village public affairs. Even when both age groups use digital technologies, younger residents demonstrate higher usage intensity and stronger ability to interpret and apply online information. As a result, the influence of digital technologies on participation is more pronounced for younger rural residents.

TABLE 11 Heterogeneity analysis of the impact of village social foundations on rural residents' participation in digital governance.

Variables	(1)	(2)	(3)	(4)	(5)	(6)
	Different age groups		Different education levels		Different political identities	
	Age ≤40	Age > 40	College and above	High school and below	Party members	Non-Party members
Village social foundations	1.12*** (0.2)	0.521*** (0.199)	1.086*** (0.251)	0.761*** (0.164)	1.437*** (0.322)	0.746*** (0.158)
Control variables	Controlled	Controlled	Controlled	Controlled	Controlled	Controlled
Observations	884	1,079	536	1,427	245	1718
R^2	0.092	0.121	0.09	0.137	0.153	0.126

4.5.2 Residents with different education levels

The regression results indicate that village social foundations exert a stronger influence on digital governance participation among highly educated residents than among those with lower levels of education. Individuals with higher education typically possess stronger digital skills and higher usage frequency, making them more likely to participate actively in digital village governance.

4.5.3 Residents with different political identities

Digital governance participation among rural residents is often reflected in activities such as digital Party-building (e.g., remote Party education platforms, "Xuexi Qiangguo") and digital administrative services (e.g., WeChat government service accounts, Alipay-based administrative services). Compared with ordinary residents, Party members tend to perform better in both the variety and frequency of digital applications they use. Therefore, the impact of village social foundations on digital governance participation is more significant among Party members.

5 Conclusion and discussion

5.1 Main conclusion

Digital rural governance is fundamentally part of state capacity building in rural society, and rural residents' participation in digital governance reflects the effectiveness of such capacity construction. Drawing on State-Society interaction theory and using survey data from 11 provinces in China, this study constructs a measure of village social foundations—comprising cultural foundations, organizational foundations, and human-capital foundations—and employs an ordered probit model to examine their effects on rural residents' participation in digital governance and the possible mechanisms involved. The findings show that strong village social foundations significantly enhance participation in digital governance, with village organizational foundations exerting the strongest influence. Village social foundations also display significant heterogeneous effects across different groups. Compared with older residents, those with lower education, and non-Party members, the effects are stronger among younger residents, more educated residents, and Party members. This suggests that, in promoting smart agriculture, targeted subsidies, and e-commerce market access, these groups should serve as "pioneers" and "diffusion agents" to advance sustainable rural development. Village social foundations primarily influence participation through three internal mechanisms: strengthening village trust, enhancing village publicness, and reinforcing Party–mass relations. These mechanisms act as the social lubricants that mitigate core challenges in sustainable food systems, such as technology adoption constraints, obstacles in policy implementation, and weak linkages between smallholders and large markets. First, strong social foundations increase village trust, promote mutual support and communication, and facilitate participation in governance. Second, by enhancing village publicness, they heighten residents' sense of responsibility and mission, thereby increasing engagement in digital governance. Third, strong social foundations reinforce Party–mass linkage mechanisms, enabling Party members to play a leading role, strengthening organizational trust, and increasing overall participation in digital governance.

5.2 Policy recommendations

First, mobilize existing cultural resources to stimulate endogenous motivation. Local cultural norms—such as village regulations, customary practices, and community-based values—can strengthen village trust and cohesion, raise residents' attention to village affairs, and create conditions for greater participation in digital governance.

Second, reinforce grassroots organizational capacity to provide institutional support for digital governance participation. Stronger grassroots Party organizations and intermediary bodies—such as elderly associations and rural mediation committees—can enhance coordination and improve residents' engagement with digital governance practices.

Third, consolidate the village's human-capital foundations, strengthen professional talent development, narrow the rural–urban digital divide, and improve rural digital literacy. More training and awareness programs are needed to enhance digital acceptance among residents with different education levels, and to reshape perceptions of digital governance. At the same time, rural mutual-help norms should be leveraged to support older residents by improving their digital adaptability. Age-friendly digital governance tools should be developed to increase inclusiveness and participation.

5.3 Discussion

Digital rural development is essentially the construction of foundational state power within rural society—a process through

which state power penetrates local communities. This necessarily involves the broader debate on State-Society relations. The complexity of state organization and the diversity of governance challenges highlight the importance of incorporating social factors into state governance; the state and society are interdependent and mutually embedded. From this perspective, the participation of rural residents in digital governance must align with village social foundations, which form the key conditions for successful digital transformation. Digital rural development is not an external technological infusion but an internal process of social activation. The findings of this study underscore that successful digital governance requires alignment with local social capital, organizational resources, and human capital. This "social-foundation adaptive" model of digital development emphasizes the integration of technological logic with social logic. It affects not only governance efficiency but also agricultural technology adoption, the precision of digital subsidies, market access for smallholders, and household resilience to risks—core elements of sustainable food systems. Without systematic support for marginalized groups' digital capacities and network accessibility, digital public services and e-government reforms may fail to enhance participation and may even entrench existing inequalities (Djatmiko et al., 2025). A village with high trust, strong publicness, and capable organizations can more effectively convert digital tools into collective action—such as adopting environmentally friendly farming practices, ensuring equitable distribution of subsidies, establishing traceable agri-food brands, and ultimately strengthening food security and livelihood resilience. Therefore, digital rural development must match technological solutions with foundational village characteristics—local culture, organizational capacity, and resident attributes. This alignment enhances the applicability and effectiveness of digital technologies and increases residents' willingness to participate in digital governance, thereby contributing to effective rural governance and sustainable agricultural development. However, challenges remain. In increasingly aging and hollowed-out rural communities, how can village publicness be revitalized? How can residents' agency in digital governance be strengthened? What types of incentive-compatible digital governance mechanisms can generate tangible social and economic rewards for sustainable food practices (e.g., waste sorting, organic farming)? How can digital governance be integrated with traditional community wisdom to support sustainable food systems? These questions point to important directions for future research. Despite offering valuable empirical evidence, this study has several limitations. First, it relies on cross-sectional data, which restricts causal inference. Although theoretical reasoning supports the positive effect of village social foundations on digital governance participation, reverse causality or unobserved confounders cannot be fully eliminated. Second, although the data cover 11 provinces, China's vast regional diversity means that our sample may not fully capture deep regional heterogeneity—for example, between economically advanced coastal villages and traditional agricultural villages in western provinces. This limits the generalizability of the findings.

Future research may therefore proceed in the following ways. Longitudinal panel designs: Tracking the same villages over time could reveal the dynamic causal relationship between social foundations and digital governance participation. Comparative regional studies: Comparative case studies across different village types (e.g., peri-urban integrated villages, traditional farming villages, culturally distinct villages) could enrich understanding of the mechanisms and boundary conditions through which village social foundations operate.

Data availability statement

The original contributions presented in the study are included in the article/supplementary material, further inquiries can be directed to the corresponding author.

Author contributions

ZX: Conceptualization, Writing – original draft, Writing – review & editing. GZ: Formal analysis, Writing – original draft, Writing – review & editing. CW: Data curation, Formal analysis, Validation, Writing – original draft, Writing – review & editing. XZ: Methodology, Formal analysis, Data curation, Validation.

Funding

The author(s) declared that financial support was received for this work and/or its publication. This research received financial support from the Ministry of Education of China Humanities and Social Sciences Research Project, "Digital Platform Construction and the Adaptive Challenges and Mechanisms of Effective Rural Governance" (Grant No. 24YJC840039). We sincerely appreciate this funding support.

Conflict of interest

The author(s) declared that this work was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

Generative AI statement

The author(s) declared that Generative AI was not used in the creation of this manuscript.

Any alternative text (alt text) provided alongside figures in this article has been generated by Frontiers with the support of artificial intelligence and reasonable efforts have been made to ensure accuracy, including review by the authors wherever possible. If you identify any issues, please contact us.

Publisher's note

All claims expressed in this article are solely those of the authors and do not necessarily represent those of their affiliated organizations, or those of the publisher, the editors and the reviewers. Any product that may be evaluated in this article, or claim that may be made by its manufacturer, is not guaranteed or endorsed by the publisher.

References

Aker, J. C., and Mbiti, I. M. (2010). Mobile phones and economic development in Africa. *J. Econ. Perspect.* 24, 207–232. doi: 10.1257/jep.24.3.207

ANTARA. (2025). Around 3,000 Indonesian villages still without signal: Komdigi. Available online at: https://en.antaranews.com/news/391665/around-3000-indonesian-villages-still-without-signal-komdigi (Accessed November 28, 2025).

Bai, X., and Yang, L. (2025). Digital literacy's impact on digital village participation in rural left-behind women through serial mediation of political trust and self-efficacy. *Sci. Rep.* 15:34226. doi: 10.1038/s41598-025-16093-3

Camacho, A., and Conover, E. (2019). The impact of receiving SMS price and weather information on small-scale farmers in Colombia. *World Dev.* 123:104596. doi: 10.1016/j.worlddev.2019.06.020

Chinese Academy of Sciences Knowledge Service Platform. (2025). The Digitalization Status of Agriculture in the European Union. Available online at: https://www.las.ac.cn/front/product/detail?id=4d5b6a355a4c0ec142e6f70efac85e81 (Accessed November 28, 2025).

Djatmiko, G. H., Sinaga, O., and Pawirosumarto, S. (2025). Digital transformation and social inclusion in public services: A qualitative analysis of E-government adoption for marginalized communities in sustainable governance. *Sustainability* 17:2908. doi: 10.3390/su17072908

Du, P. (2020). Frontline governance: mechanisms of rural governance modernization. *Polit. Sci. Res.* 4, 106–118.

Evans, P. B. (1997). State-society Synergy: Government and Social Capital in Development: University of California Press.

Fabregas, R., Kremer, M., and Schilbach, F. (2019). Realizing the potential of digital development: the case of agricultural advice. *Science* 366:eaay3038. doi: 10.1126/science.aay3038

Feng, C. (2021). Embedding village publicness in rural environmental governance. *J. China Agric. Univ. (Soc. Sci. Ed.)* 6, 69–80.

Han, P. Y. (2021). State power construction in rural society: practice and reflection. *Academics* 8, 89–98.

Hu, B. L. (2004). Village trust and rotating-savings associations. *Econ. Res. J.* 10, 115–125.

Hu, W. B., and Huang, Z. H. (2022). Collective property rights reform and village trust. *J. Zhejiang Univ. (Humanit. Soc. Sci.)* 8, 28–46.

Isabella, I., Agustian, E., Baharuddin, T., and Ibrahim, A. H. H. (2025). Bridging e-government with digital literacy: a literature review. *J. Gov. Regul.* 14, 361–371. doi: 10.22495/jgrv14i1siart12

Klerkx, L., Jakku, E., and Labarthe, P. (2019). A review of social science on digital agriculture, smart farming and agriculture 4.0. *NJAS Wageningen J. Life Sci.*, 90–91. doi: 10.1016/j.njas.2019.100315

Li, Y. P. (2020). Grassroots party-building, governance, and village publicness. *J. Nanjing Agric. Univ. (Soc. Sci.)* 2, 59–68.

Li, H., and Song, B. F. (2022). Social foundations and mobilization logic of village relocation. *J. Beijing Univ. Technol. (Soc. Sci.).* 6, 26–38.

Liang, S. M. (2011). Theory of Rural Reconstruction. Shanghai: Shanghai People's Publishing House.

Liu, S. J. (2022). Causes and solutions of "floating" digital rural construction. *J. China Agric. Univ. (Soc. Sci. Ed.).* 5, 5–12.

Liu, S. J., and Lin, A. S. (2021). The evolutionary path and experience of China's rural construction actions. *Soc. Dev. Stud.* 8, 13–22.

Liu, N., and Lu, B. Z. (2022). Fit and adaptation: practical logic of digital governance in rural society—a case study of Deqing. *J. China Agric. Univ.* 5, 25–41.

Liu, S. J., and Luo, Y. B. (2022). Effective paths for promoting digital rural construction. *Fujian Forum* 2, 182–189.

Lu, J. H., and Wei, X. D. (2021). Governing the elderly digital divide: framework, concepts and pathways. *Popul. Res.* 3, 17–30.

Ma, L., and Zhang, G. L. (2020). Internet "+" and rural governance: coupling, challenges and optimization. *E-Governance* 12, 31–39.

Mann, M. (2007). The Sources of Social Power (C. Haihong et al., Trans.). Shanghai: Shanghai People's Publishing House. (Original work published 1986)

Migdal, J. (2013). State in Society: How States and Societies Transform and Constitute One Another (L. Yang et al., Trans.). Jiangsu: Jiangsu People's Publishing House.

Prior, D. D., Mazanov, J., Meacheam, D., Heaslip, G., and Hanson, J. (2016). Attitude, digital literacy and self-efficacy: flow-on effects for online learning behavior. *Internet High. Educ.* 29, 91–97. doi: 10.1016/j.iheduc.2016.01.001

Qiu, Z. Q., Li, Y. J., and Xu, W. T. (2022). Digitalization and changes in rural governance structures. *J. Xi'an Jiaotong Univ.* 2, 74–84.

Shen, Y. D., and Bi, H. R. (2021). Multi-mechanisms of digital governance platforms in enhancing state-society co-governance. *Comp. Econ. Soc. Syst.* 6, 113–121.

Shen, F. W., and Du, F. (2022). Limits and optimization of digital rural governance. *J. Nanjing Agric. Univ. (Soc. Sci.).* 4, 134–144.

Su, L. L., and Peng, Y. L. (2021). Drivers of farmers' participation in digital rural practices. *J. Huazhong Agric. Univ. (Soc. Sci.).* 5, 168–179.

Su, L. L., and Peng, Y. L. (2022). Farmers' digital literacy, rural elites, and digital governance participation. *Agric. Technol. Econ.* 1, 34–50.

Toyama, K. (2015). Geek Heresy: Rescuing Social Change from the Cult of Technology. New York: PublicAffairs.

UNESCO (2025). Release of the global-scale sustainable development science monitoring report (2025): a decade of Progress from the perspective of earth big data. Available online at: at:https://www.unesco-hist.org/index.php?r=article/info&id=2471(Accessed November 28, 2025).

Wang, S. G. (2007). Village trust, relational communities and rural informal finance. *China Rural Surv.* 4, 75–79.

Wang, S. B. (2022). Reproduction of social foundations in rural revitalization. *Dongyue Tribune* 1, 169–175.

Zhang, H., and He, R. (2022). A study on digital inclusion of Chinese rural older adults from a life course perspective. *Front. Public Health* 10:974998. doi: 10.3389/fpubh.2022.974998

Zhang, C. E., Li, Y. M., and Li, H. (2016). Capital endowment and farmers' participation in social governance: evidence from 1,599 households in five provinces. *China Rural Surv.* 1, 27–37.

Zhao, X. F., Ma, R., and Zhao, X. Y. (2022). Social foundations of rural governance communities. *J. Beijing Univ. Technol. (Soc. Sci.).* 22, 75–84.

Zhao, G., and Sun, H. (2025). Impact of digital village development on farmers' human settlement environment improvement behaviors. *Front. Sustain. Food Syst.* 8:1526399. doi: 10.3389/fsufs.2024.1526399

Zhu, Z. Y., and Liu, C. C. (2022). The impact and threshold effects of digital infrastructure on the urban–rural income gap. *J South China Agri Univ* 5, 126–140.